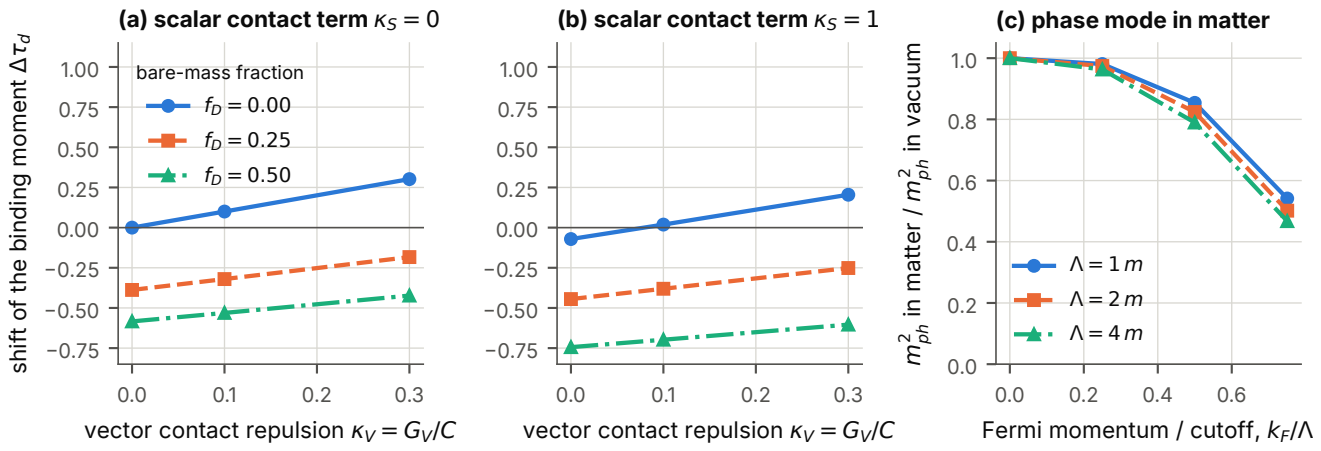
(a) scalar contact term $\kappa_S = 0$
(b) scalar contact term $\kappa_S = 1$
(c) phase mode in matter
shift of the binding moment $\Delta\tau_d$
bare-mass fraction
$f_D = 0.00$
$f_D = 0.25$
$f_D = 0.50$
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
0.0
0.1
0.2
0.3
vector contact repulsion $\kappa_V = G_V/C$
m^2_{ph} in matter / m^2_{ph} in vacuum
$\Lambda = 1\,m$
$\Lambda = 2\,m$
$\Lambda = 4\,m$
0.4
0.6
Fermi momentum / cutoff, k_F/Λ